Hospital Readmission Prediction Using Machine Learning and Deep Learning

Final Capstone Report

Diabetes 130-US Hospitals Dataset • Structured Clinical Readmission Modeling

This report integrates the complete project workflow: data wrangling, exploratory data analysis, preprocessing, model development, model comparison, and final recommendations.

Executive Summary

This project develops predictive models to identify patients at risk of 30-day hospital readmission using the Diabetes 130-US Hospitals dataset. The analysis follows a complete data science pipeline including data wrangling, exploratory data analysis (EDA), preprocessing, and model development. The dataset presents several analytically important challenges, including substantial class imbalance, heterogeneous feature types, and nontrivial missing-value patterns. These properties make simple accuracy-based evaluation inadequate and require closer attention to recall, precision–recall tradeoffs, and ROC-AUC.

Four models were evaluated: Logistic Regression, Random Forest, Gradient Boosting, and a Neural Network. Gradient Boosting achieved the highest ROC-AUC at 0.677 on the finalized notebook output, outperforming Random Forest at 0.667, Logistic Regression at 0.637, and the Neural Network at 0.625. However, Logistic Regression achieved by far the strongest recall at 0.561, compared with 0.004 for Random Forest, 0.008 for Gradient Boosting, and effectively near zero for the Neural Network. These results show that the model with the strongest ranking ability is not the model that identifies the truest readmissions at the default threshold. The final conclusion is therefore twofold: Gradient Boosting is the best model for ranking patient risk, while Logistic Regression is more useful when the primary objective is minority-class capture. The current best model should be interpreted as a risk stratification tool rather than a deployment-ready alert system.

1. Problem Statement

Hospital readmissions within 30 days are clinically and operationally important because they often indicate incomplete recovery, inadequate discharge planning, poor follow-up, medication issues, or broader care coordination problems. Readmissions also increase cost and consume limited hospital resources. From a healthcare management perspective, the value of a predictive model is not merely academic; it lies in the ability to identify high-risk patients early enough to intervene through discharge planning, follow-up scheduling, care-management outreach, or targeted review.

The objective of this project is to build and compare predictive models that estimate whether a patient encounter will result in readmission within 30 days. A secondary objective is to interpret the statistical behavior of the models under severe class imbalance and determine what type of model is most useful under different operational goals, such as ranking risk versus maximizing positive-case detection.

2. Data Description

The analysis uses the Diabetes 130-US Hospitals dataset from the UCI Machine Learning Repository. The raw dataset contains 101,766 hospital encounters and 50 original variables spanning demographics, diagnoses, medications, and encounter-level utilization characteristics such as inpatient visits, emergency visits, outpatient visits, time in hospital, and number of diagnoses.

The original readmission field contains three levels: 'NO', '>30', and '<30'. For this project, the response variable was converted into a binary outcome, `readmitted_30d`, where 1 indicates readmission within 30 days and 0 indicates otherwise. After dropping identifiers and the original

multiclass target, the modeling dataset contains 47 predictors. The positive class rate is approximately 11%, making this a strongly imbalanced classification problem and directly influencing both the choice of models and the choice of evaluation metrics.

3. Data Wrangling and Preprocessing

The wrangling stage began by converting placeholder missing values marked as '?' into proper null values. This was necessary because several clinically relevant categorical fields use placeholders rather than native missing values. Identifier variables such as encounter_id and patient_nbr were removed to prevent leakage from non-predictive administrative identifiers. The original readmission variable was then recoded into the binary target used throughout the project.

For EDA, a wrangled base dataset was saved so that the statistical structure of the data could be examined before full encoding. For modeling, categorical features were transformed into one-hot encoded variables, while numerical features were retained in continuous form. Scaling was applied to Logistic Regression and the Neural Network because those models are sensitive to feature magnitude, whereas Random Forest and Gradient Boosting were trained on the unscaled representation. The train/test split was stratified to preserve class balance across both subsets.

4. Exploratory Data Analysis

4.1 Class Imbalance

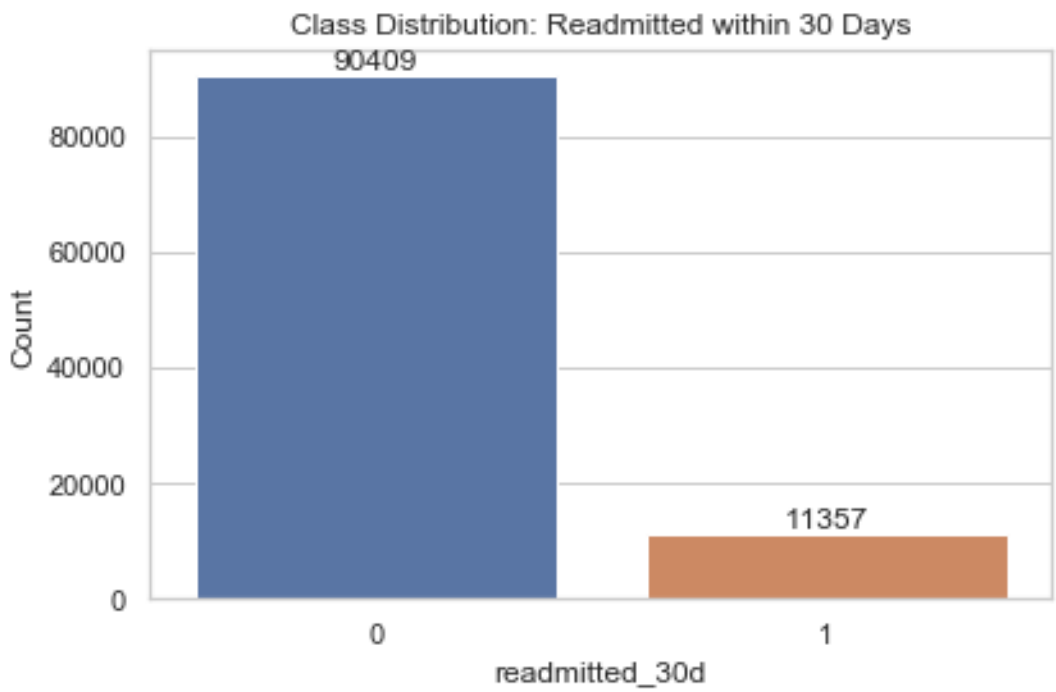

Figure 1. Class distribution for 30-day readmission.

Figure 1 shows that the dataset is strongly imbalanced: the vast majority of encounters do not lead to 30-day readmission, while only a much smaller portion belong to the positive class. This is the

defining statistical property of the problem. A classifier can achieve high accuracy simply by predicting the majority class most of the time yet still be clinically unhelpful because it misses the very patients most in need of intervention.

This figure therefore explains why accuracy cannot be treated as the primary success criterion in this project. Instead, recall, F1 score, ROC-AUC, and precision–recall behavior must be used to understand whether a model is actually capturing readmission risk.

4.2 Missing-Value Structure

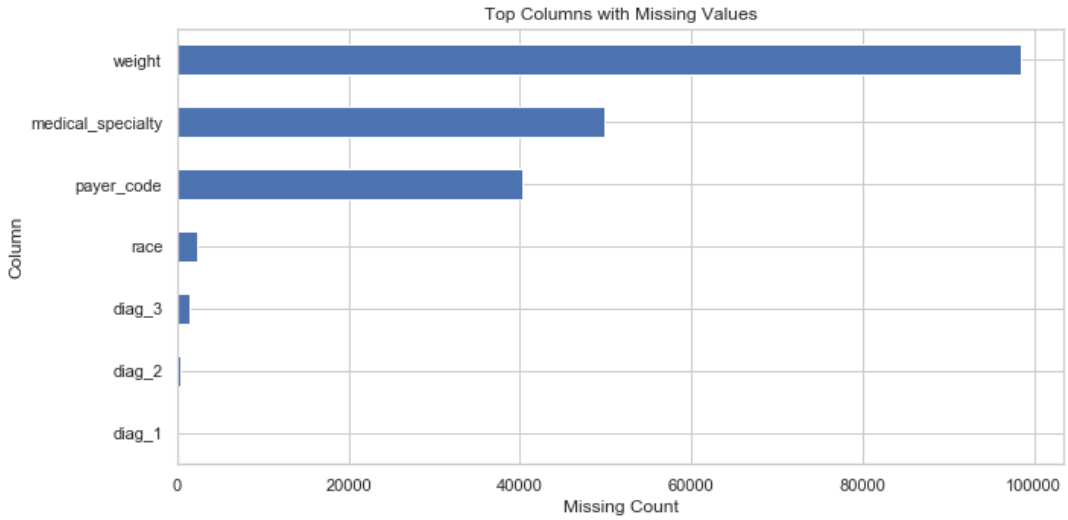

Figure 2. Top columns with missing-value markers.

Figure 2 highlights the variables with the largest number of placeholder missing values. Weight, medical specialty, and payer code dominate the missingness profile. This matters because missingness in healthcare data is not always random. It may reflect workflow priorities, documentation differences, or patient subgroup effects.

From a modeling standpoint, this figure justifies the structured preprocessing pipeline. Without systematic handling of missing values, the model would either treat placeholders as valid categories or fail during training. The figure also suggests a future avenue for improvement: missingness-aware encoding or feature engineering may increase signal extraction from incompletely observed variables.

4.3 Distribution of Core Utilization Features

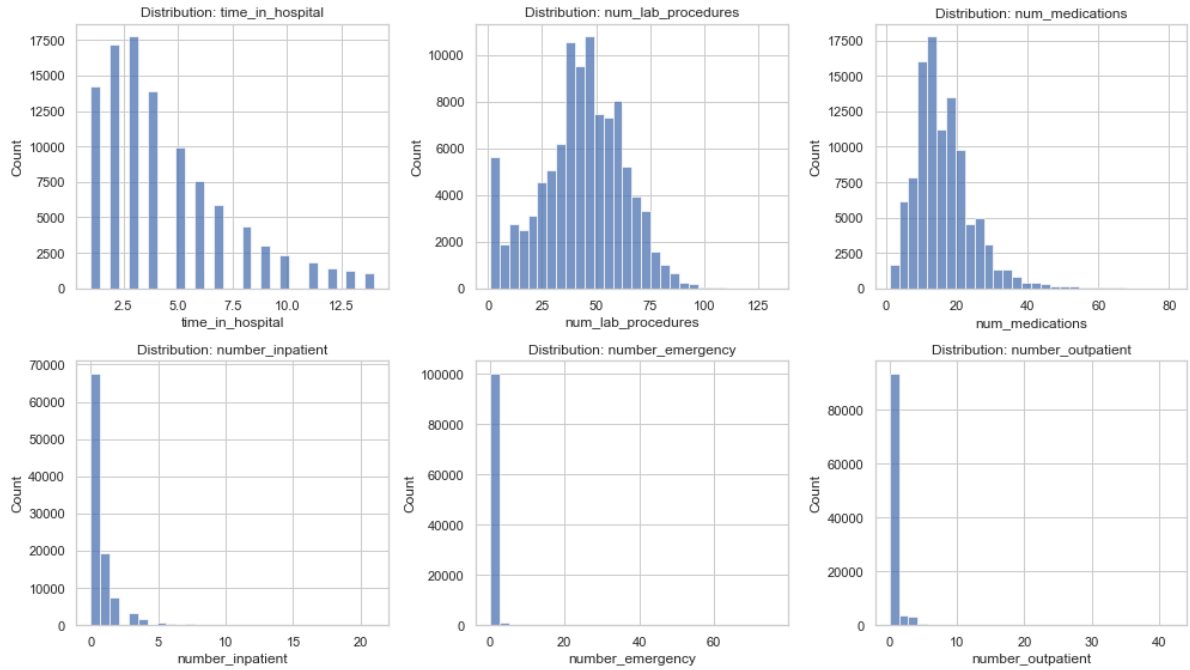

Figure 3. Distributions of key numerical utilization variables.

Figure 3 shows the distributions of several utilization-related numerical features. These distributions are heavily right-skewed: most encounters cluster at relatively low values, while a smaller subset shows very high utilization. This is consistent with real healthcare systems, where a small number of medically complex patients generate disproportionately high service use.

These patterns are scientifically informative because they suggest that extreme values are not noise but potentially signal-rich cases. Variables such as inpatient visits, emergency visits, lab procedures, and medication counts likely proxy for disease burden, acuity, or instability and therefore merit strong consideration in model interpretation.

4.4 Feature Relationships with Readmission

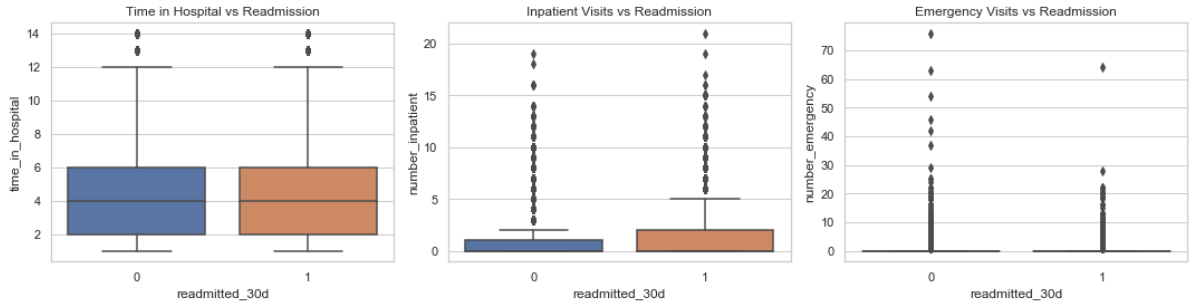

Figure 4. Utilization features versus readmission outcome.

Figure 4 compares utilization-related variables between readmitted and non-readmitted encounters. Patients in the readmitted class tend to show higher inpatient and emergency use, as well as longer

hospital stays. This result is clinically plausible, because repeated utilization often reflects unresolved illness, chronic instability, or insufficient outpatient support.

These plots are particularly important because they connect the EDA directly to the later model results. They help explain why utilization history is likely to be a central predictor of readmission risk and why a model with good nonlinear ranking ability can still perform meaningfully even when pairwise linear correlations are modest.

4.5 Age and Readmission Risk

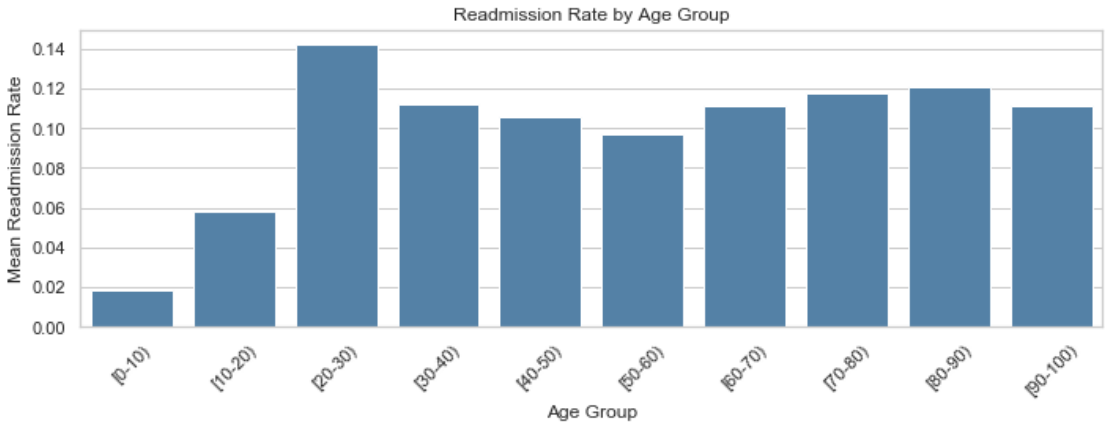

Figure 5. Readmission rate by age group.

Figure 5 shows how readmission risk varies across age groups. Although age alone is not the strongest feature, the figure indicates a tendency for higher readmission risk in older groups. This aligns with clinical intuition: older patients often have greater comorbidity burden, more complex medication patterns, and slower recovery trajectories.

Scientifically, this figure supports treating age as a meaningful baseline risk factor that likely becomes more informative when combined with utilization and diagnostic variables.

4.6 Correlation Structure

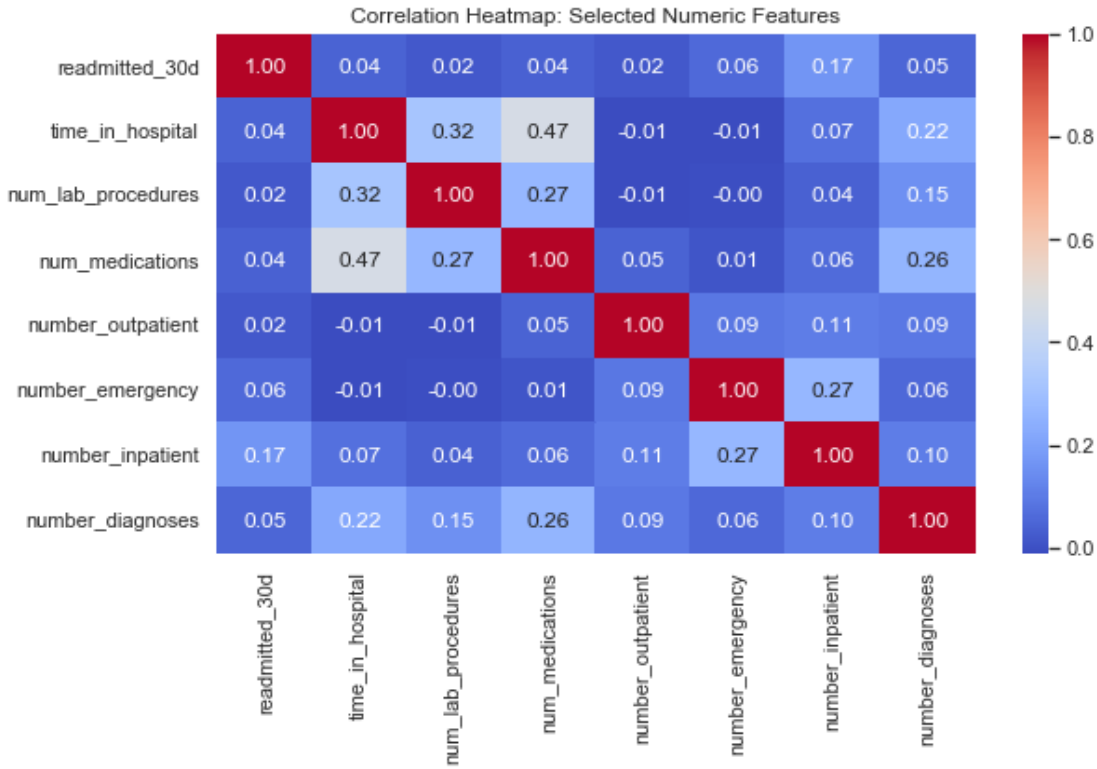

Figure 6. Correlation heatmap for selected numerical features.

Figure 6 summarizes the linear correlation structure among selected numerical predictors and the target. Most associations with the readmission outcome are modest in magnitude, and there is no single overwhelmingly dominant linear feature. This is an important analytical result because it suggests that the readmission signal is distributed across multiple variables and may arise through interactions rather than simple one-variable effects.

The heatmap therefore provides a scientific reason to compare linear and nonlinear models. Weak pairwise correlations do not imply lack of predictive value; rather, they suggest that ensemble models may be better positioned to exploit nonlinear combinations of moderately informative variables.

4.7 Outlier Review

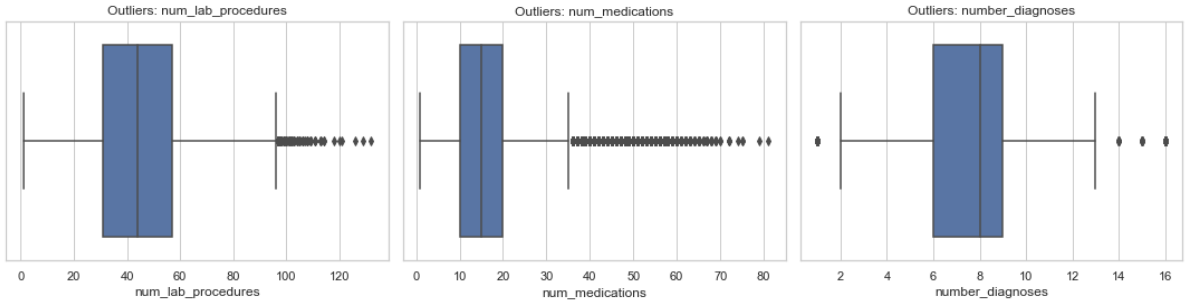

Figure 7. Outlier review for utilization-related numerical variables.

Figure 7 shows that outliers are present in several numerical variables such as lab procedures, medications, and number of diagnoses. In a clinical dataset, these extreme values are often meaningful rather than erroneous. They may correspond to medically complex patients with repeated care needs or extended diagnostic workups.

The analytical implication is that outliers should not automatically be trimmed without domain justification. In this project, they are better interpreted as part of the readmission signal rather than as noise to be discarded.

5. Modeling Approach

Four models were implemented to compare different learning assumptions. Logistic Regression served as the interpretable baseline and provided a useful benchmark for how much of the readmission signal can be captured with a linear decision boundary. Random Forest was used as a nonlinear ensemble baseline capable of capturing feature interactions across the large, encoded feature space. Gradient Boosting was evaluated because boosting often improves ranking performance on structured tabular data. Finally, a feedforward Neural Network was implemented as the advanced-model extension to test whether additional nonlinear representation learning improves predictive performance.

All models were evaluated using the same train/test split and compared using accuracy, precision, recall, F1 score, and ROC-AUC. Because the project's central scientific challenge is imbalanced classification, interpretation focuses primarily on recall and ranking performance rather than raw accuracy alone.

6. Model Performance Comparison

The table below summarizes the exact test-set performance for the classical models and the finalized neural network output. The comparison reveals that different models are optimal under different definitions of success.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.635796	0.165583	0.560546	0.255648	0.637273
Random Forest	0.888523	0.562500	0.003963	0.007871	0.667406
Gradient Boosting	0.888621	0.562500	0.007926	0.015632	0.677459
Neural Network	0.888376	0.333333	0.000440	0.000880	0.624596

The table reveals the central tradeoff in the project. Logistic Regression captures far more true readmissions than the other models, as reflected in its recall above 0.56, but it does so at the cost of

lower precision and lower overall accuracy. Random Forest and Gradient Boosting produce much higher overall accuracy and better ROC-AUC, but their recall collapses because they are extremely conservative about predicting the minority class. The Neural Network performs comparably in overall accuracy but does not improve the clinically important minority-class behavior.

7. Results and Interpretation

7.1 Logistic Regression

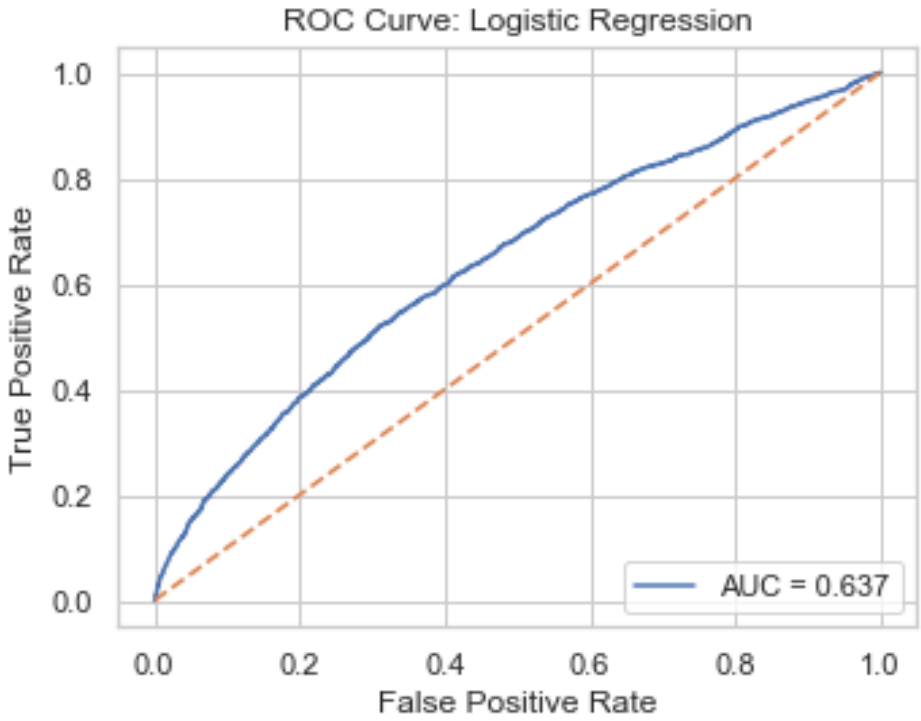

Figure 8. ROC curve for Logistic Regression.

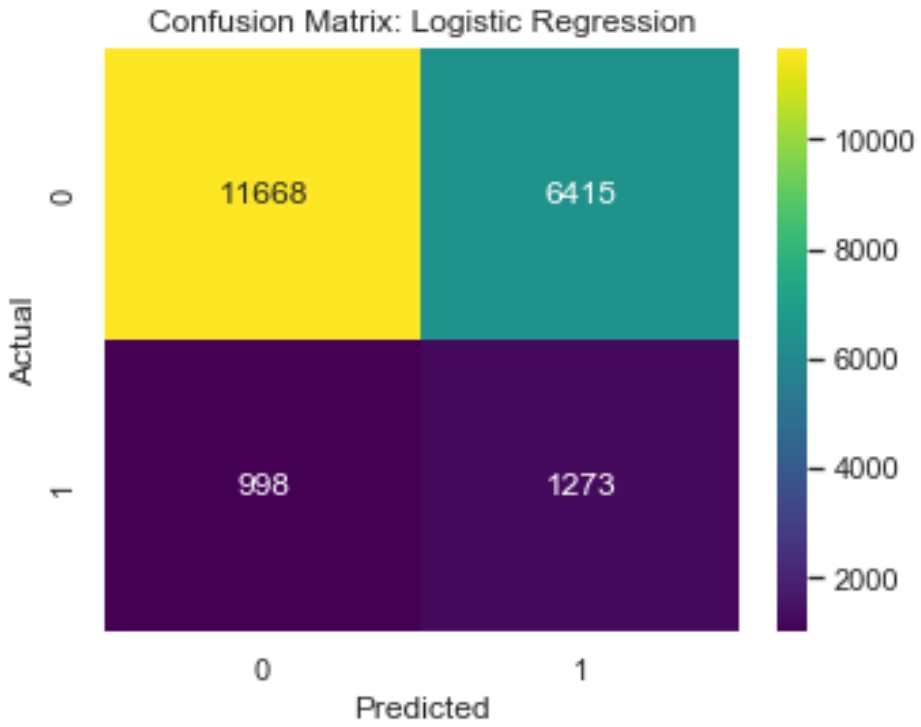

Figure 9. Confusion matrix for Logistic Regression.

Logistic Regression demonstrates the most recall-friendly behavior among the evaluated models. Its confusion matrix shows that it is willing to classify many more cases as positive, which increases sensitivity to true readmissions but also produces a large number of false positives. Scientifically, this is exactly what one would expect from a class-balanced linear model applied to an imbalanced healthcare problem.

7.2 Random Forest

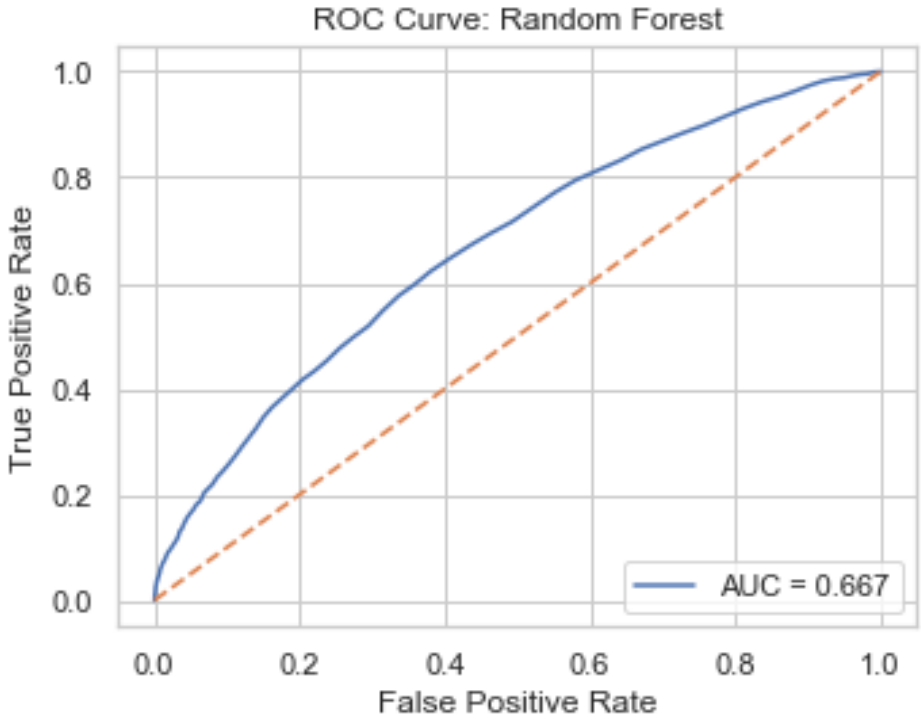

Figure 10. ROC curve for Random Forest.

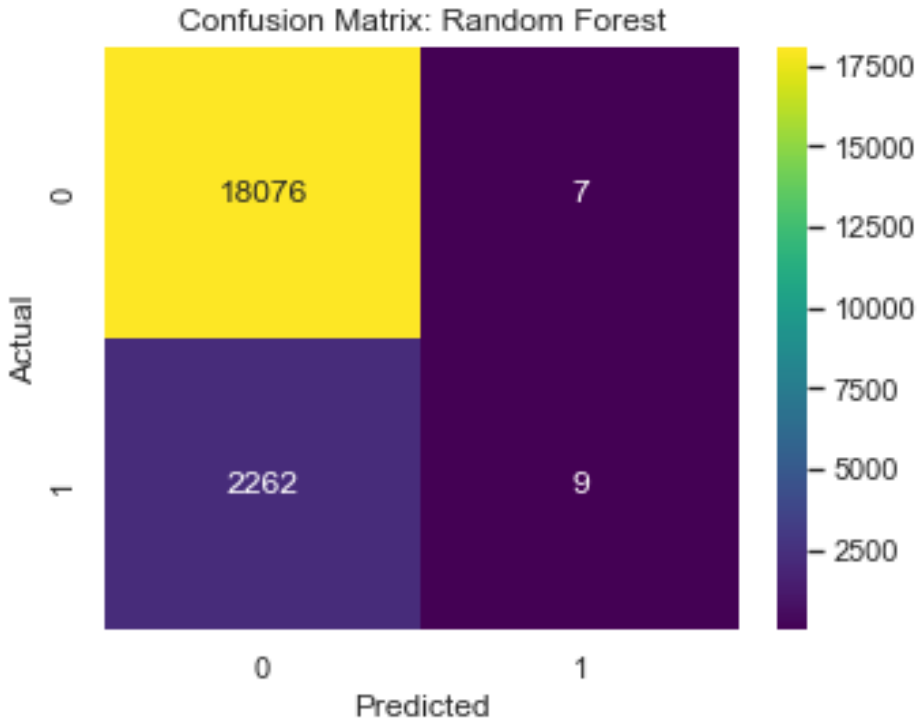

Figure 11. Confusion matrix for Random Forest.

Random Forest improves ranking performance over Logistic Regression, as shown by its higher ROC-AUC, but becomes extremely conservative at the default threshold. The confusion matrix

shows that it rarely predicts the positive class. This behavior yields high overall accuracy because the majority class dominates, but it undermines the model's usefulness for direct readmission detection.

7.3 Gradient Boosting

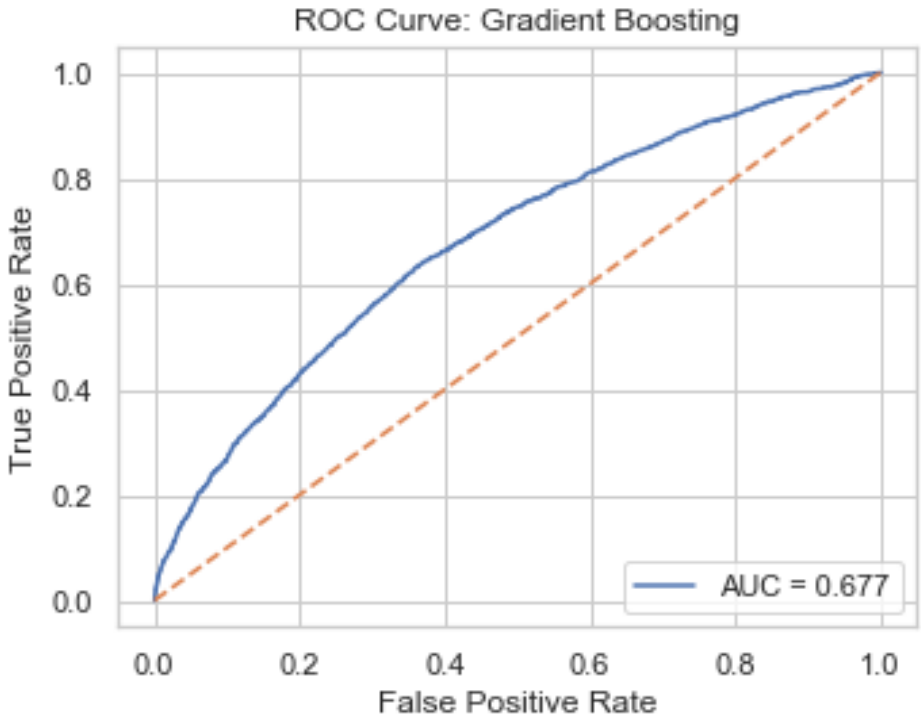

Figure 12. ROC curve for Gradient Boosting.

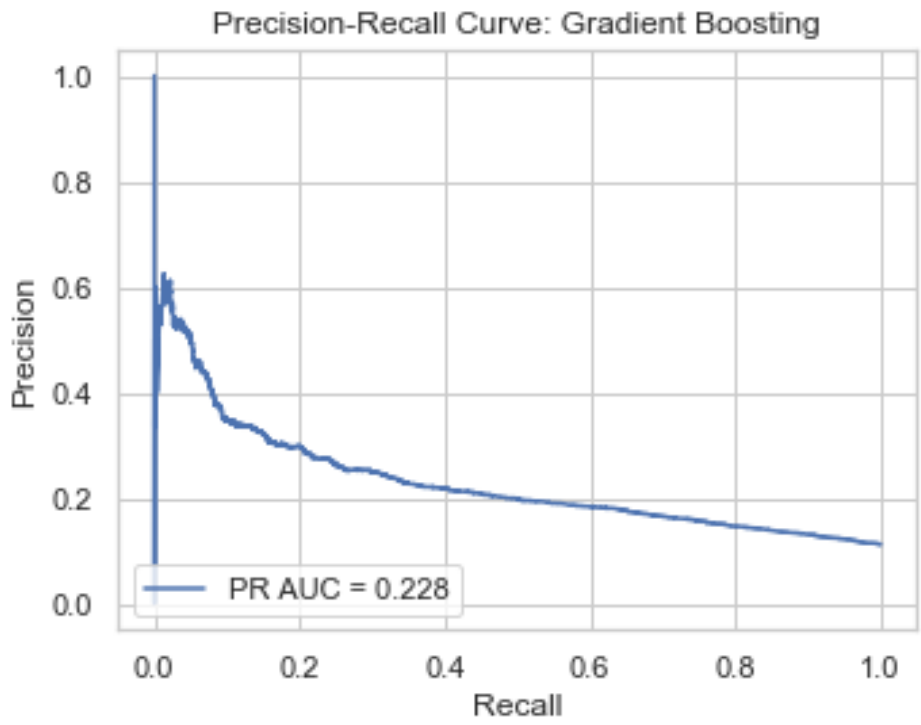

Figure 13. Precision–Recall curve for Gradient Boosting.

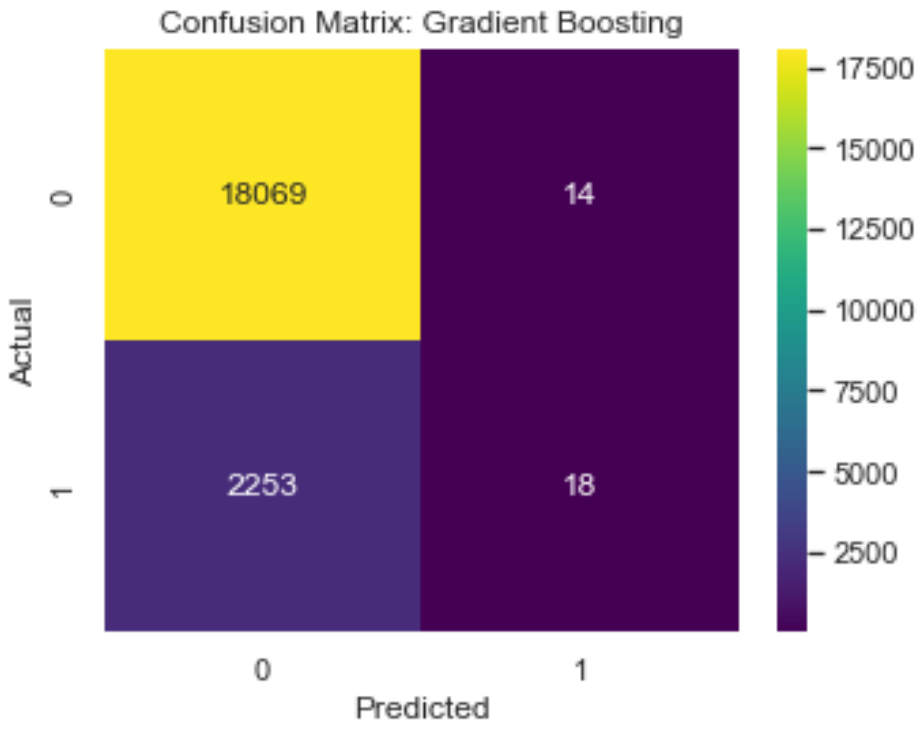

Figure 14. Confusion matrix for Gradient Boosting.

Gradient Boosting achieves the strongest overall ranking performance, as indicated by the highest ROC-AUC among the evaluated models. The ROC curve confirms that the model contains useful discriminatory structure across thresholds. However, the precision–recall curve and confusion matrix show that this ranking strength does not translate into acceptable minority-class capture at the default threshold. This is the key scientific conclusion of the project: Gradient Boosting is the best model for ranking risk, but not for default-threshold alerting.

7.4 Neural Network

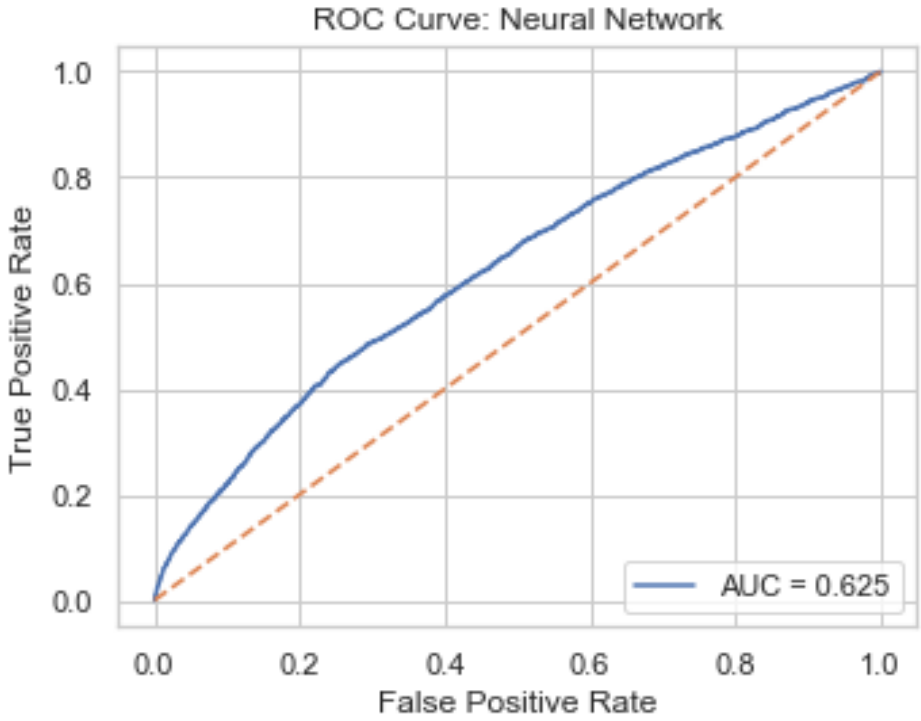

Figure 15. ROC curve for the Neural Network.

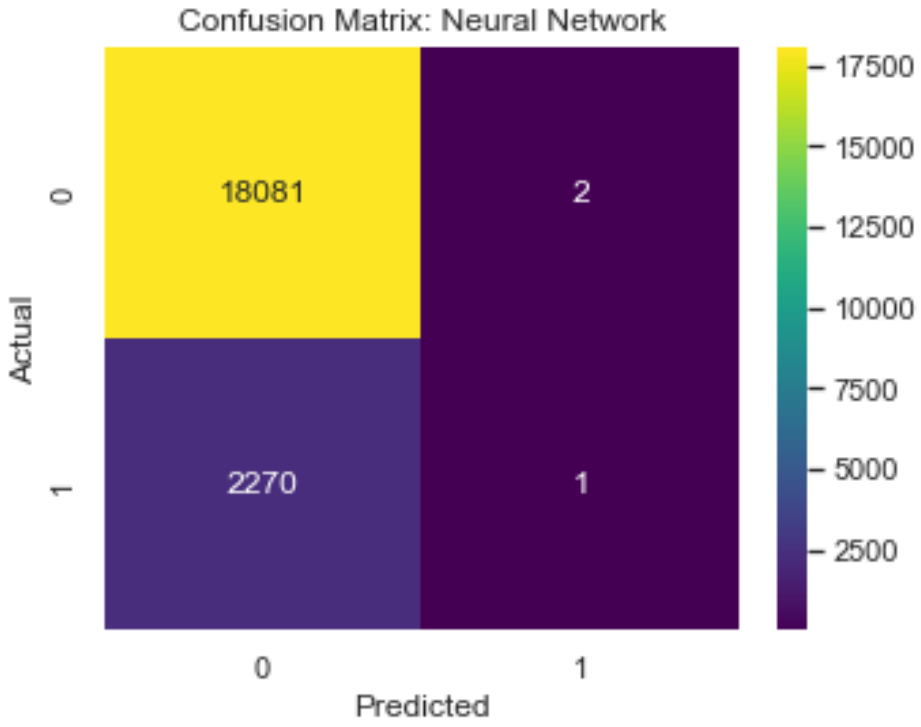

Figure 16. Confusion matrix for the Neural Network.

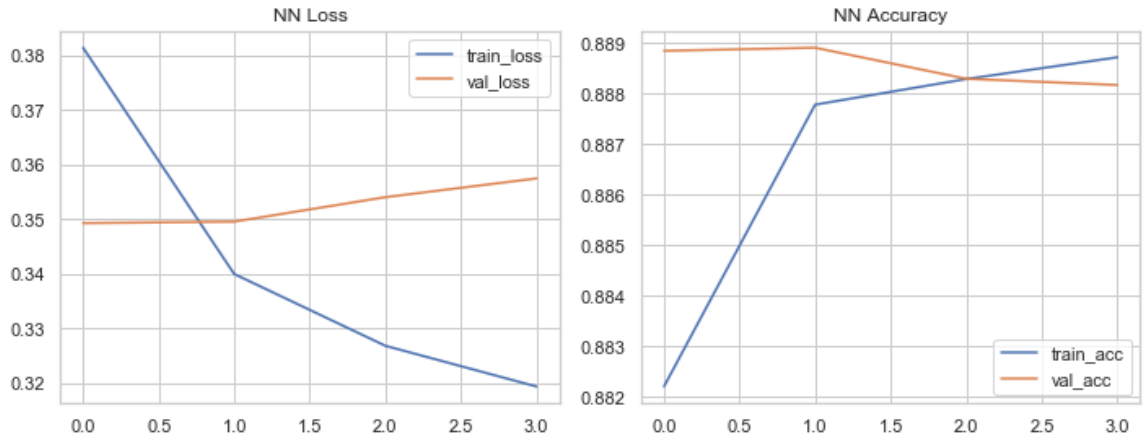

Figure 17. Neural Network training and validation history.

The Neural Network was included as the project's deep learning component. Its ROC curve indicates that it learned some useful signal, but its ROC-AUC remained below both Random Forest and Gradient Boosting. The confusion matrix shows that, like the ensemble methods, it rarely predicts the minority class at the default threshold. The training history suggests that the model converged without major instability, but that optimization alone was not enough to overcome the underlying class imbalance problem.

This result is scientifically informative. It suggests that for structured tabular hospital data, deeper nonlinear representation does not automatically outperform well-tuned ensemble trees. The Neural Network adds methodological breadth to the project and aligns with the proposal's deep learning goal, but the empirical evidence does not support choosing it as the final practical model.

8. Final Model Selection

Gradient Boosting is selected as the final model because it provides the strongest overall ranking performance, as reflected in its ROC-AUC of 0.677459. This makes it the best candidate when the use case is risk ordering or triage. However, the model's recall remains extremely low at the default threshold, meaning it cannot yet be used as a stand-alone binary readmission alert.

Logistic Regression remains valuable because it demonstrates an alternative operating regime: lower ranking performance but substantially stronger minority-class capture. Together, these findings show that model selection must be driven by the intended clinical use case rather than by a single metric alone.

9. Limitations and Further Research

This project is limited to structured variables available in the public dataset and does not include richer temporal history, free-text notes, medication adherence data, or post-discharge follow-up information. The strong class imbalance also makes threshold-dependent classification unstable and reduces the practical usefulness of default settings. Although the Neural Network provides a

meaningful advanced extension, the tabular nature of the dataset limits how much benefit can be expected from deep learning without more expressive inputs.

Further research should focus on threshold tuning, calibration, cost-sensitive learning, resampling strategies, and additional feature engineering around utilization history, diagnosis burden, and missingness patterns. It would also be valuable to incorporate richer temporal patient histories to test whether sequence-aware modeling improves minority-class detection.

10. Recommendations

First, the current Gradient Boosting model should be used as a risk-ranking system rather than a direct binary decision rule. Patients with the highest predicted probabilities can be prioritized for discharge review, follow-up outreach, and transitional care interventions. Second, threshold optimization should be performed before deployment so that the balance between false positives and missed cases aligns with the hospital's intervention capacity and tolerance for missed high-risk patients. Third, future feature engineering should focus on utilization history, diagnostic burden, and structured missingness, because both the EDA and the model comparison suggest that these are the most influential components of readmission risk.